

Solar Photovoltaic Systems: A Technical Primer

1. How PV Cells Work

A photovoltaic cell converts sunlight directly into electricity through the photovoltaic effect. Most cells are made from silicon semiconductor material; when photons strike the cell, they excite electrons and produce a flow of direct current, or DC, electricity.

2. Panel Efficiency

Panel efficiency describes how much sunlight a panel converts into usable electricity. Monocrystalline panels typically operate at 20-22% efficiency, making them the most space-efficient option. Polycrystalline panels are less expensive but operate at a lower 15-17% efficiency.

3. Inverter Types

Three inverter architectures are common in residential installations: string inverters, microinverters, and power optimizers. A microinverter converts DC to AC at each individual panel, which improves performance when panels are partially shaded.

4. System Sizing

A typical residential solar system size ranges from 5kW to 10kW, depending on household electricity consumption and available roof area. Larger systems are common in regions with high electricity prices or high air-conditioning load.

5. Maintenance

Solar panels should be inspected twice a year for physical damage and loose connections. Dust and debris accumulation can reduce panel output by up to 25% if panels are not periodically cleaned, particularly in dry or high-pollen climates.

6. Degradation Rate

Most manufacturers guarantee that panels retain at least 80% of their original rated output after 25 years of use. This corresponds to an average annual degradation rate of about 0.5% per year under normal operating conditions.